

Grade 6 Accelerated
Unit 10
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

Noelle kept track of the number of pages she read each day for the past 20 days. Her data are shown.

24, 35, 28, 27, 33, 36, 154, 28, 27, 25, 32, 25, 27, 31, 30, 28, 32, 25, 28, 30

1. One value in Noelle's data set is an outlier. Which value appears to be an outlier? Explain your reasoning.

154 because most values are in the 20's and 30's, so it's a more than 100 page difference.

2. Complete the statements.

The mean of Noelle's full data set is 35.25. $\frac{705}{20} = 35.25$

The mean of Noelle's data set with the outlier removed is 29. $\frac{551}{19} = 29$

3. Is there a significant difference between the mean of the full data set and that of the modified data set?

There is a 6.25 page difference which is somewhat significant.

4. Complete the statements.

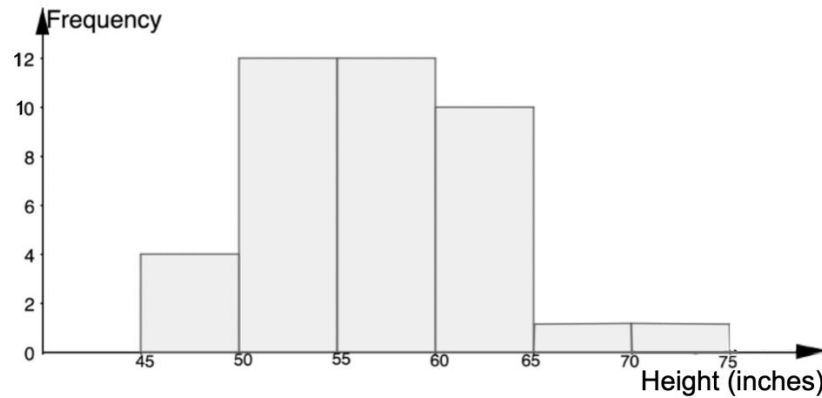
The IQR of Noelle's full data set is 5. $32 - 27$

The IQR of Noelle's data set with the outlier removed is 5. $32 - 27$

5. Is there a significant difference between the median of the full data set and that of the modified data set?

There was no difference; both are 28.

The histogram shows the heights, in inches, of students in a 6th grade class.



6. Describe the shape of the histogram.

Slightly right skewed

7. Explain which measure of center, mean or median, best represents this data.

Median because the graph is skewed

8. Explain which measure of variability, range or IQR, best represents this data.

IQR because the graph is skewed

The length of the songs on an album, in minutes, is shown.

3.5, 3.5, 3.75, 4, 4, 4, 4.25, 4.5, 5, 5, 5.5, 5.75

9. Explain which measure of center, mean or median, best represents this data.

The data appears to be symmetrical (no outliers); the mean better represents the data.

10. Explain which measure of variability, range or IQR, best represents this data.

Range because of the symmetrical data